

Deepak Mallampati

MACHINE LEARNING ENGINEER - LLM & PERSONALIZATION

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati | github.com/deepak-glitch |
US-based, F-1 OPT · open to NYC relocation

PROFESSIONAL SUMMARY

Applied AI / ML Engineer with 4+ years building and deploying production LLM and retrieval systems. Specializes in **grounding generative models on structured knowledge** - dense retrieval, cross-encoder re-ranking, and hybrid search - with RAG pipelines achieving **92% precision** and **40% hallucination reduction**. Ships production ML at scale (100k+ requests/day) with rigorous evaluation, cost, and latency optimization, and builds in AI-native workflows (Claude Code, Cursor).

CORE COMPETENCIES & SKILLS

LLMs & RAG · Agent Workflows · Embeddings & Vector Search · Recommendation / Ranking · Production ML Deployment · Model Evaluation · Prompt & Tool Orchestration · Cost & Latency Optimization

LLMs & GenAI: GPT-4o/GPT-5, Claude, Gemini, LLaMA · HuggingFace Transformers · LangChain/LangGraph · RAG · Prompt Engineering · Multi-Agent Systems · Fine-tuning (LoRA/QLoRA)

Retrieval & Ranking: FAISS, Pinecone, Weaviate · Embeddings · Semantic & Hybrid Search · Cross-Encoder Re-ranking · Semantic Caching

ML & Frameworks: PyTorch, TensorFlow · RAGAS & BERTScore Evaluation · A/B Testing · Drift Detection

MLOps & Cloud: AWS (S3, Lambda, Bedrock, SageMaker) · Docker, Kubernetes · MLflow, Weights & Biases · Model Monitoring

Data Engineering: Spark, Databricks · Kafka · PostgreSQL, Redis · SQL · Scheduled Batch Orchestration

Languages & Tools: Python, TypeScript, SQL · FastAPI · Claude Code, Cursor · Git

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present

AI Engineer

USA

- **Grounded LLM outputs on structured data** via RAG - dense retrieval + cross-encoder re-ranking + hybrid search - the core pattern behind reliable, explainable generative AI over knowledge graphs and entity stores.
- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents - reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **retrieval-augmented generation (RAG)** pipelines over large-scale healthcare document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and materially reduce hallucination.
- Established a structured **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) with latency/accuracy dashboards to benchmark iterations pre-deployment and surface regressions before release.
- Designed **privacy-preserving data workflows** for sensitive clinical datasets (K-anonymity, L-diversity, differential privacy) enabling compliant analytics with zero PII exposure.
- Optimized inference cost/latency via prompt compression, semantic caching, and model routing, sustaining SLA targets while reducing per-query overhead.

Vanguard

Jan 2024 - Jan 2025

AI Engineer Intern

USA

- Developed **AI agents and backend services** for Vanguard's internal AI platform across **25+ AI agents** and workflows, integrating with production data pipelines and observability tooling.
- Optimized prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini**, improving task success rates by **27%** and informing platform model-selection decisions.
- Built **12 APIs and microservices**, reducing p95 latency from 1.5s to **800ms** and supporting **100,000+ requests daily** across 20+ internal teams.
- Engineered content-safety safeguards for policy-compliant AI responses, reducing unsafe outputs by **42%**.

Kent State University

Oct 2023 - Jan 2024

ML & Gen AI Research Assistant

USA

- Built a **privacy-preserving clinical ML pipeline** on a hospital-readmission dataset (k-anonymity, l-diversity, Laplace differential privacy), reducing re-identification risk from **3.38% to 0.32%** while retaining **99% of baseline predictive performance**.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 for controllable long-form generation; authored an IEEE-format conference paper.

Emerson

Jun 2022 - Apr 2023

ML Trainee

India

- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification, achieving **89% F1-score** on custom NER tasks.
- Built supervised regression/classification models for equipment-failure prediction and anomaly detection with end-to-end data preprocessing.

PROJECTS

career-ops

Autonomous AI-driven pipeline that scans listings, scores fit, and generates tailored applications overnight - built on Claude Code / agentic tooling.

Privacy-Preserving Clinical ML Pipeline

Framework combining k-anonymity, l-diversity, and differential privacy to analyze sensitive patient data without exposing identities - quantifies privacy-utility trade-offs on real clinical data.

MangaLens

Shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation across 29+ sites.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025 · GPA 3.70/4.0